財政學 HW6
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1. (Hyman 13.4) Suppose leisure is an inferior good for a worker. Set up this worker's indifference curves for money income and leisure, and derive the income and substitution effects of a tax-induced wage decline. Derive the compensated labor supply curve for this worker, and explain how it differs from the compensated supply curve of a worker for whom leisure is a normal good.

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If leisure were an inferior good, both the income and substitution effects of a tax-induced wage decline would be favorable to leisure. The compensated labor supply curve would be steeper than the regular labor supply curve.

2. (Hyman 13.5) Empirical studies show that the market labor supply of prime-age males (between the ages of 25 and 55) in the United States is close to perfectly inelastic with respect to the labor compensation. If this is the case, explain why it does not imply that the excess burden of a tax on the labor income of prime-age males is necessarily zero. What does a perfectly inelastic market labor supply imply about the incidence of taxes on labor income? Why is there reason to believe that the overall market labor supply is not perfectly elastic when groups other than prime-age males are considered? What possible problems exist in empirical studies of the response to taxes on labor income that might make it difficult to estimate actual labor market responses?

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The excess burden is not zero because a perfectly inelastic market labor supply means that the income effect is offset by an equal and opposite substitution effect. The existence of a substitution effect means that there is a positive excess burden. A perfectly inelastic market labor supply implies that the incidence of the tax is borne entirely by the workers, market equilibrium wages do not rise in response to the tax, and none of the tax is shifted to the employers. Overall market labor supply includes responses of many demographic groups. Spouses and the elderly often have more elastic labor supply than prime-age males. Factors that might affect labor supply that are difficult to measure include early retirement, absenteeism, reduced intensity of work, unwillingness to invest in more training, and choice of occupation.

3. (Rosen 18.5) According to Feldstein[2008b], "only about 10% to 20%" of a one-time tax rebate provided to taxpayers in 2008 was spent immediately. Using the life-cycle model of Figure 18.7, explain why this result could have been predicted.(Hint: Think about how a one-time rebate affects the endowment point and budget constraint in Figure 18.7. Contrast this to how a permanent rebate affects the endowment point and budget constraint.)

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試想一個如 Figure 18.7，橫軸為當期消費，縱軸為未來消費的圖。

一次性退稅：

理性的個人知道一次性退稅不是常態事件，故一次性退稅只是暫時增加了個人的所得，預算限制線暫時向外移動，部分人的短期內消費可能增加。

永久性退稅：

永久性退稅增加了個人的所得，預算限制線永久向外移動，稟賦點隨預算限制線外移而可以用更多消費。

Feldstein 提到一次性退稅只有 10%到 20%會被立即消費，是因為人們預期政府未來會多課稅，故在現在多儲蓄→李嘉圖等值定理

4. (Rosen 18.10) In an economy, the supply curve of labor, S, is given by

$$S = -100 + 200W_n$$

Where W_n is the after-tax wage rate. Assume that the before-tax wage rate is fixed at 10.

a. Write a formula for tax revenue as a function of the tax rate, and sketch function in a diagram with the tax rate on the horizontal axis and tax revenues on the vertical axis. [Hint: Note that ,where is the tax rate, and that tax revenue are the product of hours worked, the gross wage, and the tax rate.] Suppose that the government currently imposes a tax rate of 70 percent. What advice would you give it?

b. At what tax rate are tax revenues maximized in this economy?

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a. The supply curve is given by S = -100 + 200wn. The gross wage is w = 10, and the net wage is wn = (1-t)w = (1-t)10. The difference, then, between the gross and net wage for any tax rate is w - wn = 10t. This is the tax collected per hour of work.

The tax revenue for any given hours of work is then the product of the tax collected per hour of work and the labor supply curve:

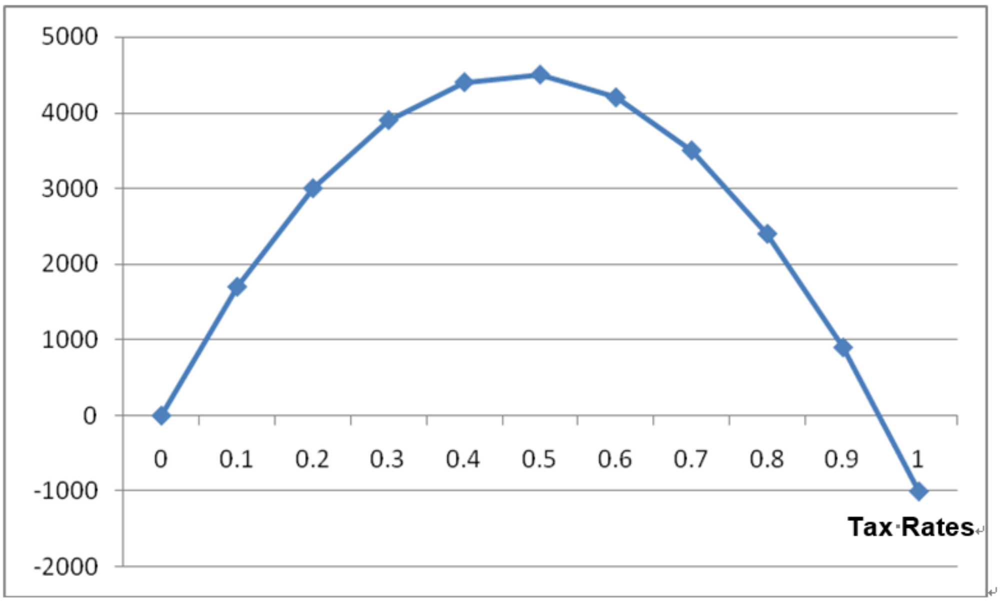
tax revenue = 10t*(-100+200(1-t)10)

$= -1{,}000t + 20{,}000t - 20{,}000t^2$

$= 19{,}000t - 20{,}000t^2$

The tax rate t = 0.7 is beyond the revenue maximizing point (this can be shown by computing tax revenue for a slightly lower tax rate, like t=0.69.

Tax Revenues



b. Taking the derivative:

dTax Revenue/dt=0

yields 19,000-40,000t=0, or

t = (19,000/40,000), or

t = 0.475 as the revenue maximizing tax rate.

5. (Rosen 19.1) Caterpillar Inc.'s CEO, Jim Owen, noted, "Sitting on a big wad of cash doesn't make any sense whatsoever for shareholders [because] it tends to promote bad practice among management…. You've got more [cash] than you know

what to do with, and you think you're so damn good you can buy anything and make it better" [Brat and Gruley, 2007]. If this view is correct, what does it imply about the efficiency consequences of the corporate tax treatment of dividends versus retained earnings?

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公司擁有大筆現金，可能助長管理層面的不良措施，造成不效率。這個觀點隱含對保留盈餘課徵公司所得稅，是能變得更有效率的做法。追求效率，會對保留盈餘課稅，公司會比以前傾向發放股利。

6. (Rosen 19.11) The ABC corporation is contemplating purchasing a new computer system that would yield a before-tax return of 30 percent. The system would depreciate at a rate of 1 percent a year. The after-tax interest rate is 8 percent, the corporation tax rate is 35 percent, and a typical shareholder of ABC has a marginal tax rate of 30 percent. Assume for simplicity that there are no depreciation allowances or investment tax credits. Do you expect ABC to buy the new computer system? Explain your answer. [Hint Use Equation (19.4)]

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The user cost of capital (ignoring depreciation allowances and investment tax credits) is given by equation (19.4) in the textbook, $C=(r+\delta)/[(1-\theta)(1-t)]$, where r=after tax rate of return in the capital market, δ=economic depreciation, θ=corporate tax rate, and t=individual tax rate. Substituting the numbers from the problem into the formula, gives the user cost: $C=(.08+.01)/[(.65)*(.7)]=.1978$. Since the project's return, 30%, is higher than this user cost of 19.8%, the company does invest in the project.